

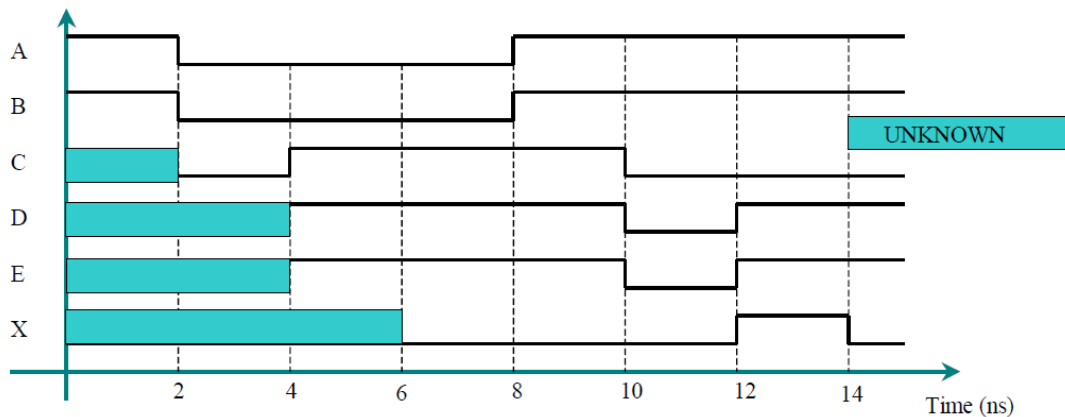
Tutorial 3: Solution

Problem 1

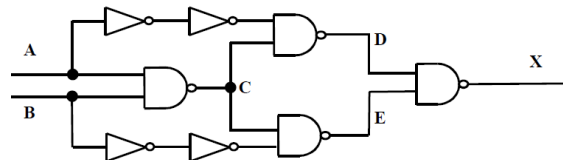
A	B	C	D	E	X
0	0	1	1	1	0
0	1	1	1	0	1
1	0	1	0	1	1
1	1	0	1	1	0

It is the exclusive or circuit.

Problem 2



Problem 3



If the inverter delay was exactly 1ns then this would be Ok, but no two inverters are identical, so there will still be a spike though much shorter than before.

Problem 4

The key observation to make is that the gate capacitances will add when we connect gates in parallel (fan-out). So, if the input capacitance for a gate is C , then in the first circuit the inverter has to supply a total capacitance of $8 \times C$. Taking the equation from lecture 5, for the change of a gates input voltage with time we have:

$$V = 5(1 - \exp(-t/RC))$$

So, if in practice we can take $V=1$ for the point where the gate has definitely switched then we can calculate the time taken to switch as:

$$5\exp(-t/RC) = 4$$

$$\exp(t/RC) = 5/4$$

$$t = RC \log(5/4)$$

so, the time taken to switch when there are eight capacitors to charge will be $t = 8RC \log(5/4)$. Now consider the second circuit. At each stage a single inverter drives just two inverters (or two nand gates at the last stage) so the time delay associated with each stage is $2RC \log(5/4)$, and so the total delay for the second circuit is: $6RC \log(5/4)$. Making the assumption that $C_{\text{nand}} = C_{\text{int}}$ (which is reasonable) the second circuit is faster.